

Air Conditioners Data Analysis

Stage 2: Data Acquisition & Filtering

This notebook is independently executable in Google Colab. Every step explains what we do, why we do it, what we obtain, and how the result is used in the next stage.

Stage Objective

Stage 2 focuses on obtaining useful subsets of the dataset through transparent filtering and sorting operations.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the project CSV directly so this notebook is independently executable.
csv_path1 = "/content/Most_Expensive_Products.csv"
df1 = pd.read_csv(csv_path1)
csv_path2 = "/content/Top_Discounted_Products.csv"
df2 = pd.read_csv(csv_path2)
csv_path3 = "/content/Top_Rated_Products.csv"
df3 = pd.read_csv(csv_path3)

print("CSV loaded successfully.")
print("Dataset shape1:", df1.shape)
print("Dataset shape2:", df2.shape)
print("Dataset shape3:", df3.shape)
```

```
CSV loaded successfully.
Dataset shape1: (15, 14)
Dataset shape2: (15, 14)
Dataset shape3: (15, 14)
```

Why Filtering Is Needed

The full product table is useful for overview, but individual analytical questions often require a focused subset. Filtering lets us isolate records without altering the original source.

```
cats=df1.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
cats=df2.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
cats=df3.select_dtypes(include=['object','category','bool']).columns.tolist(); print(cats)
```

```
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
['name', 'main_category', 'sub_category', 'image', 'link', 'brand']
```

Reload the CSV

The dataset is loaded again with `read_csv()` so this stage can be executed independently in Google Colab.

```
nums=df1.select_dtypes(include=np.number).columns.tolist(); print(nums)
nums=df2.select_dtypes(include=np.number).columns.tolist(); print(nums)
nums=df3.select_dtypes(include=np.number).columns.tolist(); print(nums)
```

```
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
['ratings', 'no_of_ratings', 'discount_price', 'actual_price', 'discount_amount', 'discount_percent', 'tonnage', 'energy_star']
```

Inspect Categories

Available categorical fields and their values are inspected before filtering. This ensures that filters use categories that actually exist in the supplied data.

```
for c in nums[:5]: print(c, df1[c].min(), df1[c].max())
for c in nums[:5]: print(c, df2[c].min(), df2[c].max())
for c in nums[:5]: print(c, df3[c].min(), df3[c].max())
```

```
ratings 1.0 5.0
no_of_ratings 1.0 18.0
discount_price 71990.0 128800.0
actual_price 77900.0 149000.0
discount_amount 2000.0 29150.0
ratings 3.0 4.6
no_of_ratings 1.0 1666.0
discount_price 199.0 38999.0
actual_price 499.0 89999.0
discount_amount 300.0 51000.0
ratings 5.0 5.0
no_of_ratings 1.0 5.0
discount_price 599.0 128800.0
actual_price 999.0 149000.0
discount_amount 400.0 29660.0
```

Inspect Numeric Ranges

Numeric fields are examined to understand their observed ranges before applying any numerical filter.

```
if cats: print(df1[cats[0]].value_counts(dropna=False).head(15))
if cats: print(df2[cats[0]].value_counts(dropna=False).head(15))
if cats: print(df3[cats[0]].value_counts(dropna=False).head(15))
```

```

name
Voltas 3Ph Scroll SAC C2-Nventure I-Cassette AC (4 Ton White) 1
Voltas Venture i Cassette 3Ph Scroll SAC(C2-N) (3 Ton) 1
Voltas Venture Slimline Tower AC (3 Ton White) 1
Blue Star 2.0 Ton 3 Star Inverter Cassette SAC (Cool Only) ID324MATURI AC (Copper, Multiple) 1
O General 2 Ton 5 Star Inverter Split AC (Copper Condenser, ASGG24CGB-B, White) 1
OGENERAL Inverter Split AC (ASGG24CGTA) 1
Mitsubishi Heavy Duty 2.2 Ton 3 Star Non Inverter Split AC (SRK25CSS, White) 1
MITSUBISHI HEAVY DUTY SRK25CSS-S6 2.2 Ton 3 Star Split Air Conditioner (White) 1
Voltas Venture Slimline Tower AC (2 Ton White) 1
Ogeneral 2 Ton 5 Star Split Inverter AC - White (ASGG24CGB, Copper Condenser) 1
O-General 2.0 Ton 5 Star Inverter Split Air Conditioner - White (ASGG24CETA-B, Copper Condenser) 1
Voltas 241 CZMM 2 TON Slimline AC, R-32 1
Ogeneral 2 Ton 3 Star Split AC (ASGA24FUTC) 1
Voltas 1 Ph Rot. SAC C2-N Venture I-Cassette AC (2 Ton White) 1
O-General 2 Ton 2 Star Split AC (ASGA24FUTC, White) 1
Name: count, dtype: int64
name
RISARYA H20R1203 IGBT 1PCS POWER TRANSISTOR 20A 1200V REVERSE CONDUCTING HEATING AND WIDELY USED IN INDUCTION COOKER T0247... 1
CNC Split AC Stand (2 Tons, Off-White) 1
LG 1.0 Ton 5 Star DUAL Inverter Wi-Fi Window AC (Copper, Convertible 4-in-1 cooling, PW-Q12WUZA, 2022 Model, HD Filter, Wh... 1
ALLWIN TAFTA High Quality Metal Special Coated Floor Mounted Split Outdoor AC Stand for 1 Ton, 1.1 Ton, 1.2 Ton, 1.5 Ton, ... 1
iMee-1.5 Ton Inverter Split Air Conditioner with Quad Air Protection -5 Star (Eco Frenldy, Cools at 52°C, Auto Clean,100% ... 1
Hexzone® TAFTA Outdoor Ac Stand (Wall) 1
Voltas 1.4 Ton 3 Star Inverter Split AC(Copper, Adjustable Cooling, Anti-dust Filter, 2023 Model, 173V Vectra Platina, White) 1
FIDEL Tafta Heavy Duty Air Conditioner Outdoor Ac stand for 1 Ton, 1.1 Ton, 1.2 Ton, 1.5 Ton, 2 Ton Outdoor Units (Pack of... 1
Voltas 1.5 Ton 3 Star, Inverter Split AC(Copper, 5-in-1 Adjustable Mode, Anti-dust Filter, 2023 Model, 183V Vertis Emeral... 1
Godrej 1.5 Ton 3 Star Inverter Split AC (AC 1.5T SIC 18TTC3-WMA, White) 1
LG 1 Ton 3 Star DUAL Inverter Split AC (Copper, Super Convertible 5-in-1 Cooling, HD Filter with Anti-Virus Protection, 20... 1
QAWACHH 3 in 1 Multi Function pet Design Mist ultrasonic Mini USB air humidifier with led Light with Mini Fan (White) 1
Motorized Electric Track Rail Slider, Dolly Rail Camera Slider Foldable with Panoramic Shooting for 4-6.4 Inch Wide Phone ... 1
Voltas 0.8 Ton 3 Star, Inverter Split AC(Copper, 4-in-1 Adjustable Mode, Anti-dust Filter, 2023 Model, 103V Vectra Elite, ... 1
Nutricook Air Fryer Mini, 1500 Watts, Digital Display, Tempered Glass Control Panel, Airfryer with 8 Preset Programs with ... 1
Name: count, dtype: int64
name
Voltas 2 Ton 5 Star Inverter Split AC (Copper Condenser, 245V EAZS, White) 1
LG 1 Ton 3 Star Split Dual Inverter AC - White (MS-Q12ENXA, Copper Condenser) 1
Daikin 1.8 Ton 5 Star Inverter Split AC (Copper,2022,FTKM60U,White) 1
Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi Inverter Smart Split AC (Copper, 7 in 1 Convertible with additional AI Mode, T... 1
AC Stand/Heavy Duty Air Conditioner Outdoor Unit Mounting Bracket 1
O-General 1 Ton 5 Star EFFICIENT & TROPICAL Inverter Split AC (Copper Condenser, ASGG12CGB-B, White) 1
Panasonic 2.2 Ton Inverter 3 Star Split Smart AC (Copper, Auto-Convertible, Twin Cool Inverter, Wifi enabled, Shield Blu A... 1
Daikin 1.5 Ton 4 Star Inverter Split AC (Copper, PM 2.5 Filter, 2022 Model, FTKN50U, White) 1
Haier Frost Clean Inverter Split AC with 5 in 1 easy convertible (1.6 Ton, 3 Star Rating, White), HSU19E-TXS3B(INV) 1
OGeneral 1.1 Ton 4 Star Eco-Friendly Window AC (White) 1
Voltas 3Ph Scroll SAC C2-Nventure I-Cassette AC (4 Ton White) 1
Whirlpool 1.5 Ton 3 Star Inverter Split AC (3D Cool Elite Pro, Copper Condenser, White) 1
Onida 1.5 Ton 3 Star Inverter Split AC (Copper IR183IDM Indium) 1
Hitachi Split Ac - 1.5 Ton Kiyora 5200Fx I Fresh Inverter - R32 - RSRG518FFEO (Gold) 1
Panasonic 2 Ton 5 Star Wi-Fi Inverter Smart Split AC (Copper, 7 in 1 Convertible with additional AI Mode, Twin Cool, PM 0.... 1
Name: count, dtype: int64

```

Category-Based View

A category-based subset is created using an actual categorical column from the dataset. The resulting row count shows how many records belong to that view.

```

if cats: filtered=df1[df1[cats[0]].notna()].copy(); print(filtered.shape)
if cats: filtered=df2[df2[cats[0]].notna()].copy(); print(filtered.shape)
if cats: filtered=df3[df3[cats[0]].notna()].copy(); print(filtered.shape)

```

```

(15, 14)
(15, 14)
(15, 14)

```

Numeric-Based View

Where numeric attributes are available, a range-based subset is demonstrated using observed values rather than arbitrary assumptions.

```

if nums: print(df1.sort_values(nums[0],ascending=False).head(10)[nums[0]])
if nums: print(df2.sort_values(nums[0],ascending=False).head(10)[nums[0]])
if nums: print(df3.sort_values(nums[0],ascending=False).head(10)[nums[0]])

```

```

0    5.0
5    5.0
4    5.0
12   4.2
3    4.0
11   4.0
1    4.0
6    3.8
8    3.7
7    3.6
Name: ratings, dtype: float64
3    4.6
2    4.2
10   4.1
1    4.0
4    4.0
6    4.0
0    4.0
13   4.0
9    4.0
12   4.0
Name: ratings, dtype: float64
0    5.0
1    5.0
2    5.0
3    5.0
4    5.0
5    5.0
6    5.0
7    5.0
8    5.0
9    5.0
Name: ratings, dtype: float64

```

✦ **Sorting**

Records are sorted using a numerical attribute so that high and low observations can be inspected directly.

```
print('Original:',len(df1),'Filtered:',len(filtered) if 'filtered' in locals() else len(df1))
print('Original:',len(df2),'Filtered:',len(filtered) if 'filtered' in locals() else len(df2))
print('Original:',len(df3),'Filtered:',len(filtered) if 'filtered' in locals() else len(df3))

Original: 15 Filtered: 15
Original: 15 Filtered: 15
Original: 15 Filtered: 15
```

✦ **Compare Row Counts**

Original and filtered row counts are compared so the effect of each filtering operation is measurable.

```
analysis_view=(filtered if 'filtered' in locals() else df1).copy(); print(analysis_view.head())
analysis_view=(filtered if 'filtered' in locals() else df2).copy(); print(analysis_view.head())
analysis_view=(filtered if 'filtered' in locals() else df3).copy(); print(analysis_view.head())
```

	name	main_category	\
0	Voltas 2 Ton 5 Star Inverter Split AC (Copper ...	appliances	
1	LG 1 Ton 3 Star Split Dual Inverter AC - White...	appliances	
2	Daikin 1.8 Ton 5 Star Inverter Split AC (Coppe...	appliances	
3	Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi In...	appliances	
4	AC Stand/Heavy Duty Air Conditioner Outdoor Un...	appliances	

	sub_category	image	\
0	Air Conditioners	https://m.media-amazon.com/images/I/31kzjzWD29...	
1	Air Conditioners	https://m.media-amazon.com/images/W/IMAGERENDE...	
2	Air Conditioners	https://m.media-amazon.com/images/I/41z1jgtpkg...	
3	Air Conditioners	https://m.media-amazon.com/images/I/71z2BRd3jb...	
4	Air Conditioners	https://m.media-amazon.com/images/W/IMAGERENDE...	

	link	ratings	no_of_ratings	\
0	https://www.amazon.in/Voltas-Inverter-Condense...	5.0	5.0	
1	https://www.amazon.in/LG-Star-Split-Dual-Inver...	5.0	5.0	
2	https://www.amazon.in/Daikin-Inverter-Split-Co...	5.0	4.0	
3	https://www.amazon.in/Panasonic-Convertible-ad...	5.0	2.0	
4	https://www.amazon.in/Stand-Conditioner-Outdoo...	5.0	2.0	

	discount_price	actual_price	discount_amount	discount_percent	tonnage	\
0	58890.0	84990.0	26100.0	30.709495	2.0	
1	35490.0	54990.0	19500.0	35.460993	1.0	
2	62250.0	83900.0	21650.0	25.804529	1.8	
3	43990.0	61400.0	17410.0	28.355049	1.5	
4	599.0	999.0	400.0	40.040040	NaN	

	energy_star	brand	
0	5.0	Voltas	
1	3.0	Lg	
2	5.0	Daikin	
3	3.0	Panasonic	
4	NaN	Ac	

	name	main_category	\
0	Voltas 2 Ton 5 Star Inverter Split AC (Copper ...	appliances	
1	LG 1 Ton 3 Star Split Dual Inverter AC - White...	appliances	
2	Daikin 1.8 Ton 5 Star Inverter Split AC (Coppe...	appliances	
3	Panasonic 1.5 Ton 3 Star Hot and Cold Wi-Fi In...	appliances	
4	AC Stand/Heavy Duty Air Conditioner Outdoor Un...	appliances	

	sub_category	image	\
0	Air Conditioners	https://m.media-amazon.com/images/I/31kzjzWD29...	
1	Air Conditioners	https://m.media-amazon.com/images/W/IMAGERENDE...	
2	Air Conditioners	https://m.media-amazon.com/images/I/41z1jgtpkg...	
3	Air Conditioners	https://m.media-amazon.com/images/I/71z2BRd3jb...	
4	Air Conditioners	https://m.media-amazon.com/images/W/IMAGERENDE...	

	link	ratings	no_of_ratings	\
0	https://www.amazon.in/Voltas-Inverter-Condense...	5.0	5.0	
1	https://www.amazon.in/LG-Star-Split-Dual-Inver...	5.0	5.0	
2	https://www.amazon.in/Daikin-Inverter-Split-Co...	5.0	4.0	
3	https://www.amazon.in/Panasonic-Convertible-ad...	5.0	2.0	
4	https://www.amazon.in/Stand-Conditioner-Outdoo...	5.0	2.0	

	discount_price	actual_price	discount_amount	discount_percent	tonnage	\
0	58890.0	84990.0	26100.0	30.709495	2.0	
1	35490.0	54990.0	19500.0	35.460993	1.0	

✦ **Create Working Views**

The useful filtered DataFrames become working views for the next stages instead of repeatedly manipulating the raw table.

```
print('Filtering stage completed.')

Filtering stage completed.
```

